Complex Analytic Geometry

From the Localization Viewpoint

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Published by

World Scientific Publishing Co. Pte. Ltd.
5 Toh Tuck Link, Singapore 596224
USA office: 27 Warren Street, Suite 401-402, Hackensack, NJ 07601
UK office: 57 Shelton Street, Covent Garden, London WC2H 9HE

Library of Congress Control Number: 2023049056

British Library Cataloguing-in-Publication Data
A catalogue record for this book is available from the British Library.

COMPLEX ANALYTIC GEOMETRY
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ISBN 978-981-4374-70-5 (hardcover)
ISBN 978-981-4374-71-2 (ebook for institutions)
ISBN 978-981-4704-29-8 (ebook for individuals)

For any available supplementary material, please visit
https://www.worldscientific.com/worldscibooks/10.1142/8324#t=suppl

Desk Editors: Nimal Koliyat/Lai Fun Kwong

Typeset by Stallion Press
Email: enquiries@stallionpress.com

Printed in Singapore

Preface

Complex analytic geometry is a subject that could be termed, in short, as the study of the sets of common zeros of complex analytic functions. It has a long history — it started with the discovery of complex numbers in the 16th century and then bloomed as the theory of analytic functions of one complex variable, including the Cauchy integral formula, in the 19th century. While the foundation for the case of several variables was laid by H. Cartan, K. Oka et al., the transcendental method was brought in algebraic geometry by G. de Rham, W.V.D. Hodge and K. Kodaira and the modern form of the subject was established in the mid-20th. At about the same time it was enriched by the works of M.F. Atiyah, R. Bott, S.-S. Chern, H. Grauert, P. Griffiths, A. Grothendieck, H. Hironaka, F. Hirzebruch, B. Malgrange, J.W. Milnor, L. Schwartz, J.-P. Serre, I.M. Singer, R. Thom, H. Whitney et al. in various related areas. This trend has been kept growing to the present time and will go on, involving many new ideas and themes. As such, the subject is closely related to many other fields of mathematics and other sciences, where numerous applications have been and will be found.

The subjects taken up in this book are rather classical, however we look at them from the localization viewpoint. This means that we are concerned with, among others, local invariants that arise naturally in complex analytic geometry and their relation with global invariants of the manifold or variety, the Poincaré-Hopf index theorem being a prototype of this. The idea is to look at them as residues associated with the localization of some characteristic classes. Two approaches are taken for this — topological and differential geometric — and the combination of the two brings out further fruitful results. For this, on the one hand we give detailed description of the Alexander duality in combinatorial topology. On the other hand we give thorough presentation of the Čech-de Rham cohomology and integration theory on it. This viewpoint provides us with a way for clearer and more precise presentations of the central concepts as well as fundamental and important results that have been treated only globally so far. It also brings new perspectives into the subject and leads to further results and applications.

Here are some key features of the book:

1. **Detailed description of the dualities and the intersection product in combinatorial topology**
   The Poincaré, Alexander and Lefschetz dualities on C^∞ manifolds are proved using tiangulations and the dual cellular decompositions. The Thom isomorphism and the Thom class are treated in this context. The intersection product is also defined combinatorially (cf. Chapter 4). These are extended to the case of singular varieties as well (cf. Chapters 13 and 14).

2. **Effective use of Čech-de Rham cohomology**
   This cohomology naturally combines the de Rham cohomology and the Čech cohomology of locally constant functions. In the global situation, it gives a canonical correspondence between the two cohomologies, which leads to the canonical de Rham theorem. The relative version is conveniently used to describe the Alexander duality, the Thom class and various localizations in terms of differential forms (cf. Chapter 7). We may also consider Čech-Dolbeault cohomology in a similar manner. In the global situation, this leads to the canonical Dolbeault theorem (cf. Section 11.4). Although it is not treated here, the relative version has many applications including the localization theory for Atiyah classes and explicit representation of Sato hyperfunctions and related operations.

3. **Topological and differential geometric definitions of characteristic classes**
   The Chern classes of a complex vector bundle and the Euler class of an oriented real vector bundle are defined topologically, via obsrtuction theory. For general Chern polynomials the corresponding classes of a complex vector bundle are defined differential geometrically, via the Chern-Weil thoery. Modifying this theory, the classes are naturally defined in the Čech-de Rham cohomology using difference forms for several connections. The relative (localized) version is also discussed in each framework (cf. Chapters 5 and 8). A direct proof is given, including the relative case, of the fact that the differential geometric Chern

class is the image of the topological one by the canonical morphism from the cohomology with integral coefficient to that with complex coefficient (cf. Section 10.5).

4. **Combination of algebra, geometry and analysis**
   The theory of analytic functions of one variable is a model of mathematical theory, where algebra, geometry and analysis are organically combined. We attempted to do similar things in the higher-dimensional case, see in particular Chapters 12 and 13.

The book starts off with basic materials on analytic functions of one or several complex variables in Chapter 1. These functions are also referred to as holomorphic functions and are the ones we are mainly concerned with. We introduce the notion of the germ of a holomorphic function and study the structure of the ring of germs of holomorphic functions. The Weierstrass preparation theorem allows us to treat the ring almost as a polynomial ring. It continues by introducing, in Chapter 2, complex manifolds and analytic varieties, the spaces we work on. We study the relation between germs of varieties and ideals in the ring of germs of holomorphic functions. Particularly noteworthy is the Nullstellensatz (zero locus theorem), which we prove in the case of principal ideals. As for the general case, we quote some structual results and give a proof based on these. We also discuss dimensions of varieties. This chapter includes some remarks on the underlying real structures as well, because of their importance.

In Chapter 3, fiber bundles, in particular vector bundles, are discussed. The homotopical structure of Stiefel manifold is studied for the obstruction theoretical definition of characteristic classes of vector bundles. The cellular structure of Grassmann manifold, which carries a unversal vector bundle, is analyzed. Some notions concerning differentiable structures are also made precise.

We deal with, in Chapter 4, the topics mentioned in the item 1 above, i.e., the Poincaré, Alexander and Lefschetz dualities, the Thom isomorphism and the intersection product in combinatorial topology. In Chapter 5, we define characteristic classes of vector bundles topologically via the obstruction theory. Their localizations by frames and the associated topological residues are also introduced. In Chapter 6, we review differential forms and their integrations. Such topics as integration along fibers and non-singular foliations are also presented.

In Chapter 7, we discuss the Čech-de Rham cohomology in detail (cf. the item 2 above). After the de Rham and Čech cohomologies are reviewed, they are combined to form the Čech-de Rham cohomology. The de Rham cocycles and the Čech cocycles of locally constant functions are naturally thought of as being Čech-de Rham cocycles and this leads to the canonical isomorphism between the de Rham cohomology and the Čech cohomology of locally constant functions. We then present the integration theory on Čech-de Rham cohomology introducing the notion of a system of honeycomb cells. Combining with combinatorial topology, we establish various canonical isomorphisms (Theorem 7.4). The canonical de Rham theorem immediately follows from this. We discuss the relative Čech-de Rham cohomology and prove the relative Čech-de Rham theorem (Corollary 7.8). It is expressed explicitly in terms of integration on topological chains and provides a bridge between topological localizations and differential geometric localizations. It also allows us to describe the dualities in terms of differential forms. The Thom class is expressed explicitly in the relative Čech-de Rham cohomology. As an application, we prove the Lefschetz fixed point formula in this context.

In Chapter 8, characteristic classes of vector bundles are defined differential geometrically via the Chern-Weil theory. The Bott difference forms are then conveniently used to modify the theory to have the classes in Čech-de Rham cohomology. This way of representing characteristic classes is particularly effective in dealing with the localization problem (cf. item 3 above). In Chapter 9, we discuss some topics related to vector bundles with metrics. Among them are harmonic forms (real and complex), Dolbeault cohomology, Kodaira-Serre duality, Hodge structures, Kähler manifolds, Atiyah classes of holomorphic vector bundles and Bott-Chern classes of Hermitian vector bundles.

We explain the fundamental idea of localizing characteristic classes in Chapter 10. We then present the residue theorem and give explicit expressions of the residues. Various types of localizations are exhibited. In particular, the differential geometric localizations of Chern classes of vector bundles by frames is discussed in detail. In Chapter 11, we present various tools and notions that are common in complex analytic geometry and are used in later chapters. In particular, we discuss complex analytic spaces, generalizations of varieties, in some detail.

In Chapter 12, we study the residues associated with localizations of Chern classes by families of sections of holomorphic vector bundles on complex manifolds. The residue is expressed explicitly (Theorem 12.2) in the case the singular set has an expected dimension, called the "proper case".

It shows that, in order to find the residue in the proper case, it suffices to know the residue at an isolated singularity. We give topological, analytic and algebraic expresions of the residue at an isolated singularity. The localized duality (Theorem 12.8) between the Chern class localized by a family of sections of a holomorphic vector bundle and the class of the complex space defined by the family of sections also follows from Theorem 12.2. It is effectively used in the localized intersection theory later in Chapter 14. In Chapter 13, we attempt to do similar things for residues of vector bundles on singular varieties. We discuss Poincaré, Alexander, Lefschetz and Thom morphisms for singular varieties in combinatorial topology. We then present de Rham and Čech-de Rham theories for singular varieties and express the above morphisms in terms of integration of differential forms. We discuss Chern classes and their localizations both from topological and differential geometric viewpoints and show that they are essentially the same. We express the residue explicitly (Theorem 13.9), as in the case of manifolds. It again shows that, in order to find the residue in the proper case, it suffices to know the residue at an isolated singularity. We give topological, analytic and algebraic expresions of the residue at an isolated singularity. We also give examples and discuss related topics. The localized duality (Theorem 13.18) between the Chern class localized by a section of a holomorphic vector bundle and the class of the complex space defined by the section also follows from Theorem 13.9.

We discuss, in Chapter 14, intersection products of the homology classes of complex subspaces in a complex manifold or in a singular variety. They are treated from the residue theoretical viewpoint and the intersection products are localized at the set of intersection. In the case of manifolds, the intersection product in general is defined combinatorially in Chapter 4, as mentioned above. It is an operation dual to the cup product in cohomology. In general, expected formulas do not hold for the intersection product of the classes of subspaces. The intersection products we consider involve, in some form or another, spaces defined by a family of sections of a vector bundle so that localizations of Chern classes and the associated residues are involved. For the residues, expected formulas always hold, thus in the proper case, for subspaces defined by a family of sections, they also hold. As to the intersection product in sigular varieties, we do not have general combinatorial ways of defining it. Thus we consider intersection products of homology classes that come naturally as the images of the Poincaré or Alexander morphism and use the cup product in cohomology. For this, as in the case of manifolds, we consider intersection product involving subspaces defined by a family of sections and develop the theory in parallel with that case. We then specialize the above considerations to the case of intersection products in singular surfaces. Finally we study the case where a variety and a complex space defined by a section intersect in a dimension greater than expected.

In Chapter 15, we discuss the Riemann-Roch theorem in the form A. Grothendieck formulated (cf. Theorem 15.11). It consists of the embedding part and the projection part. We examine the former in detail, as an application of our localization process by the exactness of vector bundle sequences, and prove a prototype of the localized Riemann-Roch theorem for embeddings on the level of Čech-de Rham cocycles (Corollary 15.1). It immediately yields a theorem in the universal situation (Theorem 15.4). We then present three variations of the localized Riemann-Roch theorem for embeddings (Theorems 15.5–15.7). For the projection part, we outline the proof quoting relevant materials. The Grothendieck-Riemann-Roch theorem contains the Hirzebruch-Riemann-Roch theorem as a special case. The latter is also generalized as the Atiyah-Singer index theorem, which we briefly recall in Notes at the end of the chapter.

The book is supplemented with two appendixes, one on commutative algebra and the other on algebraic topology.

Exercises provided in the text are meant to help better understand the materials and should mostly be done rather straightforward.

Here is a word about the use of terminologies "natural" and "canonical". A canonical morphism is the one that is induced from a natural idetification or correspondence of the objects. What is natural is somewhat a subjective matter. For example, we could embed the ring of integers $\mathbb{Z}$ into the complex numbers $\mathbb{C}$ by assigning to 1 in $\mathbb{Z}$ any non-zero complex number. In this case, we think the assignment $1 \mapsto 1$ is natural and the morphism $H_p(X;\mathbb{Z}) \to H_p(X;\mathbb{C})$ induced on the homology of a space X by this identification is canonical. As a typical example appearing in the text, Čech-de Rham cochains naturally combine de Rham cochains (differential forms) and Čech cochains of locally constant functions, as mentioned above, and the isomorphism between the de Rham cohomology and the Čech cohomology given via the Čech-de Rham cohomology is canonical.

Some symbols are used to denote different objects. For example, the letter m is used for the order of a convergent power series, the dimension of a C^∞ manifold and so forth, the letter D for a number of purposes and the notation $(f_1, \dots, f_r)$ for the map with component functions f_i, $i = 1, \dots, r$, as well as the ideal generated by the germs of the f_i's. Hopefully it will be clear from the context what it means and there will be no confusion.

Besides calculus and linear algebra, basics of the following are desirable prerequisites:

(1) Analytic functions of one complex variable (reviewed in the text as needed).
(2) Commutative algebra (reviewed in Appendix A).
(3) General topology.
(4) Algebraic topology (reviewed in Appendix B).
(5) Differentiable manifolds (reviewed in the text as needed).

There are numerous literatures on these subjects and the citations made in Notes at the end of each chapter should be supplemented with "for example".

Acknowledgements. This work was supported by JSPS Grants-in-Aid for Scientific Research Grant over the years.

I am indebted to many people through collaboration, discussion and advice in their field of expertise and it is not possible to acknowledge all of them. I owe much to J.-P. Brasselet for the item 1 above, to D. Lehmann for the item 2 and to J. Seade for the item 3. I would also like to express my appreciation to M. Abate, P. Aluffi, D. Angella, C. Bisi, F. Bracci, M. Corrêa, M. Hanamura, N. Honda, T. Izawa, K. Kurano, Y. Matsumoto, J. Noguchi, M. Oka, K. Saito, J. Schürmann, M. Shiota, M. Soares, N. Tardini, A. Tomassini, F. Tovena and S. Yokura. Special thanks are due to T. Ohmoto for his constant interest, encouragement and advice during the preparation of the manuscripts, which took way longer than initially planned. I would like to extend my gratitude to all who were involved in the editorial and production process of the book, especially Kwong Lai Fun of World Scientific and Nimal Koliyat of ACES, for their generous support.

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Fall 2023

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About the Author

Tatsuo Suwa is a Professor Emeritus at Hokkaido University, Japan. He had held academic positions at the University of Tokyo, the University of Michigan, Hokkaido University and Niigata University. He had visiting positions at institutions in a number of countries, including France, Italy, Mexico, Spain and the U.S.A. He has also been invited to give talks at symposiums and schools all over the world. He published a book on residues of singular holomorphic foliations in 1998. He then noticed that the philosophy and technics of localizing characteristic classes used there are applicable to other problems in Complex Analytic Geometry and related areas, some of which have been realized and are presented in this book.

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Contents

Chapter 1

Analytic Functions of Several Complex Variables

An analytic function is a function that can locally be expressed as a convergent power series. In the case of one complex variable, it is equivalent to saying that the function is holomorphic, meaning that it is differentiable with respect to the complex variable. As a function of the real and imaginary parts of the variable, it can also be rephrased by saying that the function is continuously differentaible and satisfies the Cauchy-Riemann equations (cf. Theorem 1.1 below).

We start by discussing convergence of multiseries, which is necessary for the case of several variables. We then review basics on analytic functions of one complex variable, notably the Cauchy integral formula, which has a significant influence throughout the book. The case of several variables follows with relation to the one variable case. The terms "analytic" and "holomorphic" will be treated as synonyms hereafter. Regular and singular points of a holomorphic map are defined and the behavior of a holomorphic map near a regular point is clarified (Theorems 1.4, 1.5 and 1.6).

The notion of the germ of a holomorphic function is introduced and the algebraic structure of the ring of germs of holomorphic functions is discussed. The Weierstrass preparation theorem (Theorems 1.10) allows us to treat convergent power series as polynomials to some extent. These are used to analyze the local structure of the set of common zeros of holomorphic functions, i.e., analytic varieties, in the next chapter.

In the following, we denote by $\mathbb{C}$ the field of complex numbers and $\mathbb{C}^n$ the product of n copies of $\mathbb{C}$. An element z in $\mathbb{C}^n$ is expressed as $z = (z_1, \ldots, z_n)$ with each z_i, $i = 1, \ldots, n$, in $\mathbb{C}$ and the set $\mathbb{C}^n$ has a natural complex vector space structure of dimension n. The norm of z is defined by

$$\|z\| = \sqrt{|z_1|^2 + \cdots + |z_n|^2},$$

which makes $\mathbb{C}^n$ a topological vector space.

Recall that, starting from the field $\mathbb{R}$ of real numbers, we may form a real vector space $\mathbb{R}^m$ of dimension m with the Euclidean norm. For a point $z = (z_1, \ldots, z_n)$ in $\mathbb{C}^n$, we write $z_i = x_i + \sqrt{-1}\, y_i$ with x_i and y_i the real and imaginary parts of z_i. Identifying z with $(x_1, y_1, \ldots, x_n, y_n)$, we may identify $\mathbb{C}^n$ with $\mathbb{R}^{2n}$ and by this identification, the above norm in $\mathbb{C}^n$ coincides with the Euclidean norm in $\mathbb{R}^{2n}$.

1.1 Multiseries

Let $\mathbb{N}_0$ denote the set of non-negative integers. Suppose a complex number c_ν is assigned to each $\nu = (\nu_1, \ldots, \nu_n)$ in $\mathbb{N}_0^n$. We call the formal infinite sum

$$\sum_\nu c_\nu = \sum_{\nu_1, \ldots, \nu_n = 0}^{\infty} c_{\nu_1 \ldots \nu_n}$$

a *multiseries*, or a *series* for short.

Definition 1.1. A series $\sum c_\nu$ *converges* to a complex number C if, for every positive number ε, there exists N in $\mathbb{N}_0$ with the following property:

$$\text{for every} \quad (N_1, \ldots, N_n) \quad \text{in} \quad \mathbb{N}_0^n \quad \text{with } N_i > N, \quad \left| \sum_{\nu_1=0}^{N_1} \cdots \sum_{\nu_n=0}^{N_n} c_\nu - C \right| < \varepsilon.$$

If this is the case, we write $C = \sum c_\nu$ and call it the sum of the series. We say that $\sum c_\nu$ is *convergent* if it converges to some complex number. Note that if a series $\sum c_\nu$ is convergent, its sum is uniquely determined.

We also say that $\sum c_\nu$ is *absolutely convergent* if the series $\sum |c_\nu|$ is convergent. From the definition we have:

Proposition 1.1. *A series $\sum c_\nu$ is absolutely convergent if and only if the set*

$$\left\{ \sum_{\nu \in \Gamma} |c_\nu| \;\middle|\; \Gamma \subset \mathbb{N}_0^n, \text{ finite} \right\}$$

is bounded. In this case the sum $\sum |c_\nu|$ is the supremum of this set.

In general, for a convergent series $\sum c_\nu$ and a finite set Γ in $\mathbb{N}_0^n$, we set

$$\sum_{\nu \notin \Gamma} c_\nu = \sum_\nu c_\nu - \sum_{\nu \in \Gamma} c_\nu.$$

Thus if a series $\sum c_\nu$ is absolutely convergent, for every positive number ε, there exists a finite set Γ such that $\sum_{\nu\notin\Gamma} |c_\nu| < \varepsilon$.

In the case $n = 1$, i.e., the case of simple series, if a series $\sum c_\nu$ is convergent, the set $\{c_\nu\}$ is bounded. Also in this case, if it is absolutely convergent, it is convergent by the Cauchy criterion. In the case $n > 1$, we may not have the boundedness of $\{c_\nu\}$, even if $\sum c_\nu$ is convergent. However we have:

Proposition 1.2. *If a series $\sum c_\nu$ is absolutely convergent, it is convergent. Moreover the sum $C = \sum c_\nu$ does not depend on the order of summation, i.e., for every bijection $\varphi : \mathbb{N}_0 \to \mathbb{N}_0^n$, the simple series $\sum_{\mu=0}^\infty c_{\varphi(\mu)}$ converges to C.*

Proof. Take a bijection $\varphi : \mathbb{N}_0 \to \mathbb{N}_0^n$. By Proposition 1.1, the simple series $\sum_{\mu=0}^\infty c_{\varphi(\mu)}$ is absolutely covergent, and thus convergent. We denote its sum by C_φ and show that $\sum c_\nu$ converges to C_φ so that $C_\varphi = C$ and it does not depend on φ. For this, let ε be an arbitrary positive number. Then there exists N_0 in $\mathbb{N}_0$ such that

$$\Big|\sum_{\mu=0}^{N_0} c_{\varphi(\mu)} - C_\varphi\Big| < \frac{\varepsilon}{2} \quad \text{and} \quad \sum_{\mu=N_0+1}^{\infty} |c_{\varphi(\mu)}| < \frac{\varepsilon}{2}.$$

Set $\Gamma_0 = \varphi(\{0,\ldots,N_0\})$ and choose N in $\mathbb{N}_0$ so that $\Gamma_0 \subset [0,N]^n$. For $(N_1,\ldots,N_n)$ with $N_i > N$, we set $\Gamma_1 = \{\,\nu \mid 0 \le \nu_i \le N_i\,\}$. Then we have

$$\Big|\sum_{\nu\in\Gamma_1} c_\nu - C_\varphi\Big| \le \Big|\sum_{\nu\in\Gamma_0} c_\nu - C_\varphi\Big| + \sum_{\nu\in\Gamma_1\smallsetminus\Gamma_0} |c_\nu| < \varepsilon.$$

□

Example 1.1. Let $z = (z_1,\ldots,z_n)$ be a point in $\mathbb{C}^n$ with $0 \ne |z_i| < 1$. For each ν in $\mathbb{N}_0^n$, we set $z^\nu = z_1^{\nu_1}\cdots z_n^{\nu_n}$. By Propositions 1.1 and 1.2, the series $\sum_\nu z^\nu$ is convergent and

$$\sum_\nu z^\nu = \frac{1}{1-z_1}\cdots\frac{1}{1-z_n}.$$

By the convention $0^0 = 1$, the above holds for every z with $|z_i| < 1$.

Next we consider series whose terms are functions. Thus let R be a subset in $\mathbb{C}^n$. Suppose we have, for each ν, a complex valued function f_ν defined on R. We denote by $\|f_\nu\|_R$ the supremum of $|f_\nu(z)|$ on R.

Definition 1.2. A series $\sum f_\nu$ is *normally convergent* on R if the series $\sum \|f_\nu\|_R$ is convergent.

Note that if a series $\sum f_\nu$ converges normally on R, the series $\sum f_\nu(z)$ converges absolutely for each z in R and it defines a function F on R:

$$F(z) = \sum_\nu f_\nu(z).$$

Proposition 1.3. *If a series $\sum f_\nu$ converges normally on R, it converges uniformly on R, i.e., we may take N in Definition 1.1 independently of z in R, with c_ν and C replaced by $f_\nu(z)$ and $F(z)$, respectively.*

Proof. Given a positive number ε. Take ε' so that $0 < \varepsilon' < \varepsilon$. By the normal convergence of $\sum f_\nu$, there exists a finite set Γ_0 in $\mathbb{N}_0^n$ such that $\sum_{\nu \notin \Gamma_0} \|f\|_R < \varepsilon'$. Take N in $\mathbb{N}_0$ so that $\Gamma_0 \subset [0, N]^n$. For $(N_1, \dots, N_n)$ with $N_i > N$, we set $\Gamma_1 = \{\, \nu \mid 0 \le \nu_i \le N_i \,\}$. Then for an arbitrary finite set Γ in N_0^n, we have

$$\Big| \sum_{\nu \in \Gamma \setminus \Gamma_1} f_\nu(z) \Big| \le \sum_{\nu \in \Gamma \setminus \Gamma_1} \|f\|_R < \varepsilon'.$$

Hence we have

$$\Big| \sum_{\nu \in \Gamma_1} f_\nu(z) - F(z) \Big| = \Big| \sum_{\nu \notin \Gamma_1} f_\nu(z) \Big| \le \varepsilon' < \varepsilon.$$

□

From the above we see that, for instance, if f_ν is continuous for each ν, the sum $F = \sum f_\nu$ is also continuous.

Series we are particularly interested in here are "power series". In the following, for $z = (z_1, \dots, z_n)$ in $\mathbb{C}^n$ and $\nu = (\nu_1, \dots, \nu_n)$ in $\mathbb{N}_0^n$, we set

$$z^\nu = z_1^{\nu_1} \cdots z_n^{\nu_n}, \quad |\nu| = \nu_1 + \cdots + \nu_n \quad \text{and} \quad \nu! = \nu_1! \cdots \nu_n!.$$

Also, for a point $a = (a_1, \dots, a_n)$ in $\mathbb{C}^n$ and an n-tuple $\rho = (\rho_1, \dots, \rho_n)$ of positive real numbers, we set

$$\Delta(a, \rho) = \{\, z \in \mathbb{C}^n \mid |z_i - a_i| < \rho_i \,\}$$

and call it the *polydisk* with center a and polyradius ρ. We may express it as a product of one-dimensional disks:

$$\Delta(a, \rho) = \Delta(a_1, \rho_1) \times \cdots \times \Delta(a_n, \rho_n).$$

For a fixed a, they form a fundamental system of neighborhoods of a.

Definition 1.3. A *power series* with center a is a series $\sum f_\nu$ such that each f_ν is of the form

$$f_\nu(z) = c_\nu (z - a)^\nu, \qquad c_\nu \in \mathbb{C}.$$

In the above, we adopt the convention as in Example 1.1. For power series, we have the following fundamental lemma:

Lemma 1.1. *If a series* $\sum c_\nu (z-a)^\nu$ *converges absolutely at* $z = w$, *the series is normally convergent on the closure* $\overline{\Delta(a,\rho)}$ *of every polydisk* $\Delta(a,\rho)$ *with* $\rho_i < |w_i - a_i|$.

Proof. Without loss of generality we may assume that $a = (0, \dots, 0)$, the origin in $\mathbb{C}^n$. For ρ as above, we may find an n-tuple $t = (t_1, \dots, t_n)$ of real numbers with $0 < t_i < 1$ and $\rho_i < t_i|w_i|$. Since the series $\sum |c_\nu w^\nu|$ is convergent, there is a constant K such that $|c_\nu w^\nu| \le K$ for all ν. If z is a point in $\overline{\Delta(0,\rho)}$,

$$|c_\nu z^\nu| \le |c_\nu|\rho^\nu \le K t^\nu$$

and the lemma follows from Example 1.1. □

Remark 1.1. As is seen in the proof, the same result holds under the assumption that $\{|c_\nu w^\nu|\}$ is bounded. Thus if $n = 1$, only the convergence of the series $\sum c_\nu (z-a)^\nu$ at $z = w$ is sufficient to have the same result.

1.2 Analytic functions of one complex variable

We recall some basics on analytic functions of one complex variable. Let U be an open set in the complex plane $\mathbb{C}$ and f a complex valued function on U.

Definition 1.4. We say that f is *analytic* at a point a in U if there is a power series $\sum_{\nu \ge 0} c_\nu (z-a)^\nu$ which converges at each point z in a neighborhood of a and satisfies

$$f(z) = \sum_{\nu \ge 0} c_\nu (z-a)^\nu$$

in a neighborhood of a. We say that f is analytic in U if it is analytic at every point in U.

In general, let r be a non-negative integer. Recall that a function of real variables is *of class* C^r, C^r for short, if its partial derivatives exist up to order r and are continuous. By convention, C^0 means continuous. If all the partial derivatives exist, we say it is C^∞.

If we write $z = x + \sqrt{-1}\,y$ with x and y the real and imaginary parts, we may think of f as above as a function of (x, y). If f is analytic in U, it is C^∞ in U.

Definition 1.5. We say that f is *holomorphic* at a point a in U if the limit

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$$

exists. We say that f is holomorphic in U if it is holomorphic at every point in U.

The above limit, if it exists, is denoted by $\frac{df}{dz}(a)$ and is called the *derivative* of f at a. If f is holomorphic in U, we may think of $\frac{df}{dz}$ as a function on U.

If f is analytic in U, it is holomorphic in U and the derivative is also analytic in U.

Cauchy integral formula: Let f be a holomorphic function on a neighborhood of the closure of a disk Δ in $\mathbb{C}$ and $\gamma = \partial\Delta$ the boundary of Δ oriented counterclockwise. Then for z in Δ,

$$f(z) = \frac{1}{2\pi\sqrt{-1}} \int_\gamma \frac{f(\zeta)\, d\zeta}{\zeta - z}. \tag{1.1}$$

Remark 1.2. The above formula holds more generally if we replace Δ with a relatively compact open set U in $\mathbb{C}$ such that $\overline{U}$ is a real two-dimensional submanifold of $\mathbb{C}$ with piecewise C^∞ boundary ∂U. In this case $\gamma = \partial U$ consists of a finite number of piecewise C^∞ simple closed curves oriented as the boundary of $\overline{U}$ (cf. Section 3.7 below).

This will be generalized later to the case f is C^∞ (Theorem 11.8 below).

For a function f on an open set U in $\mathbb{C}$, we write $f = u + \sqrt{-1}\, v$ with u and v the real and imaginary parts.

Theorem 1.1. *The following are equivalent*:

(1) *f is analytic in U.*
(2) *f is holomorphic in U.*
(3) *f is C^1 in (x, y) and satisfies the "Cauchy-Riemann equations" in U*:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Note that, if we introduce the operators

$$\frac{\partial}{\partial z} = \frac{1}{2}\Big(\frac{\partial}{\partial x} - \sqrt{-1}\,\frac{\partial}{\partial y}\Big) \quad \text{and} \quad \frac{\partial}{\partial \bar z} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + \sqrt{-1}\,\frac{\partial}{\partial y}\Big),$$

we may express the Cauchy-Riemann equations as

$$\frac{\partial f}{\partial \bar z} = 0.$$

If this is the case, we have $\frac{\partial f}{\partial z} = \frac{df}{dz}$.

In the following, we use the words "analytic" and "holomorphic" interchangeably. Here are some basic facts that we need. We denote by D a connected open set in $\mathbb{C}$.

Uniqueness of analytic continuation: Let f and g be holomorphic functions on D. If $f = g$ on a neighborhood of a point in D, $f = g$ on D.

Open mapping theorem: Let f be a non-constant holomorphic function on D. Then f is open as a map $D \to \mathbb{C}$.

Maximum principle: Let f be a holomorphic function on D. If there is a point z_0 in D such that $|f(z_0)| \geq |f(z)|$ for all z in a neighborhood of z_0, then f is constant on D. This may be thought of as a special case of the open mapping theorem.

We finish this section by reviewing the following:

Meromorphic functions

Let f and g be holomorphic functions in a neighborhood of a point a in $\mathbb{C}$. Suppose g has no zeros away from a, then f/g is defined and holomorphic away from a. We examin the behavior of f/g near a. If f is not identically equal to zero in a neighborhood of a, we may write

$$f(z) = (z-a)^k f_1(z), \qquad g(z) = (z-a)^{k'} g_1(z)$$

with k and k' non-negative integers and f_1 and g_1 holomorphic functions without zeros in a neighborhood of a. There are two cases:

(1) $k \geq k'$. In this case, we have

$$\frac{f(z)}{g(z)} = (z-a)^{k-k'} \frac{f_1(z)}{g_1(z)}$$

so that f/g extends to a holomorphic function in a neighborhood of a. If $k > k'$, the function has a zero at a. We call $m = k - k'$ the *order of the zero*.

(2) $k < k'$. In this case, we have

$$\frac{f(z)}{g(z)} = \frac{1}{(z-a)^{k'-k}} \frac{f_1(z)}{g_1(z)}.$$

We say that f/g has a *pole* at a and call $m = k' - k$ the *order of the pole*.

Definition 1.6. A *meromorphic function* φ on an open set U in $\mathbb{C}$ is a holomorphic function on the complement of a (possibly empty) set S of isolated points in U such that φ has a pole at each point of S.

Note that in a neighborhood of each point of U, φ may be written $\varphi = f/g$, where f and g are holomorphic and g is not identically equal to zero.

Remark 1.3. 1. In fact, every meromophic function φ on an open set U in $\mathbb{C}$ may be written $\varphi = f/g$, where f and g are holomorphic functions on U and g is not identically equal to zero in any connected component of U. This is a property particular to open sets in $\mathbb{C}$.

2. If we denote by O_U the set of holomorphic functions on an open set U in $\mathbb{C}$, then the usual addition and multiplication of functions give O_U the structure of a ring (in fact a $\mathbb{C}$-algebra). In particular, if $U = D$ is a non-empty connected open set, then O_D is an integral domain (cf. Section A.2), by the uniqueness of analytic continuation. Moreover, by 1 above, the set M_D of meromorphic functions on D may be thought of as the fraction field of O_D.

1.3 Analytic functions of several complex variables

Let f a complex valued function on an open set U in $\mathbb{C}^n$.

Definition 1.7. We say that f is *analytic* at a point a in U if there is a power series

$$\sum c_\nu (z-a)^\nu = \sum_{\nu_1,\dots,\nu_n \geq 0} c_{\nu_1 \dots \nu_n} (z_1 - a_1)^{\nu_1} \cdots (z_n - a_n)^{\nu_n}$$

which converges absolutely at each point z in a neighborhood of a and satisfies

$$f(z) = \sum c_\nu (z-a)^\nu$$

in a neighborhood of a. We say that f is analytic in U if it is analytic at every point in U.

Note that, in the case $n = 1$, Definitions 1.4 and 1.7 are equivalent (cf. Remark 1.1).

If f is analytic in U, it is continuous in U by Lemma 1.1 and it is analytic in each variable z_i separately, since in a neighborhood each point

a in U we may rearrange the power series as a convergent power series in $z_i - a_i$ by Proposition 1.2. In fact we have:

Theorem 1.2. *For a complex valued function f defined on an open set U in $\mathbb{C}^n$, the following conditions are equivalent*:

(1) *f is analytic in U.*
(2) *f is continuous and is analytic in each variable z_i in U, $i = 1, \ldots, n$.*

Proof. It remains to show that (2) implies (1). Let a be a point in U and choose ρ so that $\overline{\Delta(a,\rho)}$ is contained in U. For each i we denote by γ_i the boundary of $\Delta(a_i, \rho_i)$ oriented counterclockwise. By a repeated use of the Cauchy integral formula (1.1), for z in $\Delta(a,\rho)$,

$$\begin{aligned} f(z) &= \frac{1}{2\pi\sqrt{-1}} \int_{\gamma_1} \frac{f(\zeta_1, z_2, \ldots, z_n)}{\zeta_1 - z_1} d\zeta_1 \\ &= \cdots \\ &= \left(\frac{1}{2\pi\sqrt{-1}}\right)^n \int_{\gamma_1} \frac{d\zeta_1}{\zeta_1 - z_1} \cdots \int_{\gamma_n} \frac{f(\zeta_1, \ldots, \zeta_n)}{\zeta_n - z_n} d\zeta_n \\ &= \left(\frac{1}{2\pi\sqrt{-1}}\right)^n \int \cdots \int_{\gamma} \frac{f(\zeta_1, \ldots, \zeta_n)}{(\zeta_1 - z_1)\cdots(\zeta_n - z_n)} d\zeta_1 \cdots d\zeta_n, \end{aligned}$$

where the last multi-integral is to be performed on $\gamma = \gamma_1 \times \cdots \times \gamma_n$. Note that we used the continuity of f in the last equality.

On the other hand, noting that

$$\frac{1}{\zeta_i - z_i} = \frac{1}{\zeta_i - a_i} \frac{1}{1 - \frac{z_i - a_i}{\zeta_i - a_i}} \quad \text{and} \quad \left|\frac{z_i - a_i}{\zeta_i - a_i}\right| < 1,$$

we have a power series expansion

$$\frac{1}{\zeta_1 - z_1} \cdots \frac{1}{\zeta_n - z_n} = \sum_{\nu_1, \ldots, \nu_n = 0}^{\infty} \frac{(z_1 - a_1)^{\nu_1}}{(\zeta_1 - a_1)^{\nu_1 + 1}} \cdots \frac{(z_n - a_n)^{\nu_n}}{(\zeta_n - a_n)^{\nu_n + 1}},$$

which converges normally with respect to ζ on γ. Hence we may integrate term by term and we have a power series expansion, which converges absolutely at z:

$$f(z) = \sum c_\nu (z - a)^\nu$$

with

$$c_\nu = \left(\frac{1}{2\pi\sqrt{-1}}\right)^n \int \cdots \int_{\gamma} \frac{f(\zeta_1, \ldots, \zeta_n) d\zeta_1 \cdots d\zeta_n}{(\zeta_1 - z_1)^{\nu_1 + 1} \cdots (\zeta_n - z_n)^{\nu_n + 1}}.$$

□

It is known that we may remove the continuity condition in (2) above (Hartogs' theorem).

In general, let f be a complex valued function defined on an open set U in $\mathbb{C}^n$. Writing $z_i = x_i + \sqrt{-1}\,y_i$, we may think of f as a function of the real variables $(x_1, y_1, \ldots, x_n, y_n)$. We recall that (cf. Theorem 1.1) a C^1 function f is analytic in the variable z_i if and only if it satisfies the Cauchy-Riemann equation

$$\frac{\partial f}{\partial \bar{z}_i} = 0. \tag{1.2}$$

If f is analytic in U, by Lemma 1.1, it is C^1 in U, in fact it is C^∞, since we may differentiate it term by term when expressed as a power series and the resulting series also converges normally in the same domain. Thus from Theorem 1.2, we have:

Theorem 1.3. *For a function f defined on U, the following are equivalent*:

(1) *f is analytic in U.*
(2) *f is C^1 and satisfies* (1.2) *in U, for $i = 1, \ldots, n$.*

In the following, we call an analytic function also a *holomorphic* function and use the words "analytic" and "holomorphic" interchangeably as in the case of one variable.

Note that, for a holomorphic function f, the "partial derivative" $\partial f/\partial z_i$ coincides with the derivative of f considered as a holomorphic function of the single variable z_i and, for every ν, the (higher) partial derivative

$$\frac{\partial^\nu f}{\partial z^\nu} = \frac{\partial^{|\nu|} f}{\partial z_1^{\nu_1} \cdots \partial z_n^{\nu_n}}$$

exists and is holomorphic in U. If $f(z) = \sum c_\nu (z-a)^\nu$ is a power series expansion of f, then each coefficient c_ν is given by

$$c_\nu = \frac{1}{\nu!} \frac{\partial^\nu f}{\partial z^\nu}(a).$$

This series is called the *Taylor series* of f at a.

Definition 1.8. 1. Let U be an open set in $\mathbb{C}^n$ and $f : U \to \mathbb{C}^r$ a map. We say that f is *holomorphic* if, writing f componentwise as $f = (f_1, \ldots, f_r)$, each f_i is holomorphic.

2. Let U and U' be two open sets in $\mathbb{C}^n$ and $f : U \to U'$ a map. We say that f is *biholomorphic* if f is bijective and if both f and f^{-1} are holomorphic.

Proposition 1.4. *Let U and U' be open sets in $\mathbb{C}^n$ and $\mathbb{C}^r$, respectively, and let $f : U \to U'$ and $g : U' \to \mathbb{C}$ be maps. If both f and g are holomorphic, the composition $g \circ f$ is also holomorphic.*

Proof. First we note that $g \circ f$ is C^1. If we denote coordinates of $\mathbb{C}^n$ by $(z_1, \ldots, z_n)$ and those of $\mathbb{C}^r$ by $(w_1, \ldots, w_r)$, using the chain rule we have

$$\frac{\partial g \circ f}{\partial \bar{z}_i}(z) = \sum_{j=1}^{r} \frac{\partial g}{\partial w_j}(f(z)) \frac{\partial f_j}{\partial \bar{z}_i}(z) + \sum_{j=1}^{r} \frac{\partial g}{\partial \bar{w}_j}(f(z)) \frac{\partial \bar{f}_j}{\partial \bar{z}_i}(z).$$

Then we have the proposition as each term in the right-hand side is zero by the Cauchy-Riemann equations. □

For a holomorphic map f from an open set U in $\mathbb{C}^n$ to $\mathbb{C}^r$, we write $f = (f_1, \ldots, f_r)$ and set

$$\frac{\partial(f_1, \ldots, f_r)}{\partial(z_1, \ldots, z_n)} = \begin{pmatrix} \dfrac{\partial f_1}{\partial z_1} & \cdots & \dfrac{\partial f_1}{\partial z_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial f_r}{\partial z_1} & \cdots & \dfrac{\partial f_r}{\partial z_n} \end{pmatrix}.$$

We call it the *Jacobian matrix* of f with respect to z. The *rank* of f at a point a in U is the rank of the Jacobian matrix at a.

Definition 1.9. We say that a point a in U is a *regular point* of f if the rank of f at a is maximal possible, i.e., $\min(n, r)$. Otherwise we say that a is a *singular point* , or a *critical point*, of f.

Note that, if a is a regular point of f, there is a neighborhood U_a of a in U such that every point in U_a is a regular point of f.

In the case $n = r$, the determinant of the Jacobian matrix is called the *Jacobian* of f. In this case, a is a regular point of f if and only if

$$\det \frac{\partial(f_1, \ldots, f_n)}{\partial(z_1, \ldots, z_n)}(a) \neq 0.$$

If we denote by u_i and v_i the real and the imaginary parts of f_i, we have:

$$\det \frac{\partial(u_1, v_1, \ldots, u_n, v_n)}{\partial(x_1, y_1, \ldots, x_n, y_n)} = \left| \det \frac{\partial(f_1, \ldots, f_n)}{\partial(z_1, \ldots, z_n)} \right|^2. \tag{1.3}$$

Exercise 1.1. Verify the identity (1.3).

The following three theorems describe the nature of a holomorphic map near its regular point. Thus we consider maps f from a neighborhood of the origin 0 in $\mathbb{C}^n$ to $\mathbb{C}^r$. Without loss of generality, we may assume $f(0) = 0$. We express this situation by writing $f : (\mathbb{C}^n, 0) \to (\mathbb{C}^r, 0)$. Also $f : (\mathbb{C}^n, 0) \to (\mathbb{C}^n, 0)$ being biholomorphic means there exist neighborhoods U and V of 0 such that f is a biholomorphic map of U onto V.

First we prove the following using the corresponding theorem in the real variable case:

Theorem 1.4 (Inverse mapping theorem). *Let $f : (\mathbb{C}^n, 0) \to (\mathbb{C}^n, 0)$ be a holomorphic map. If 0 is a regular point of f, then f is biholomorphic.*

Proof. By (1.3) and the inverse mapping theorem in the real variable case, there exist neighborhoods U and U' of 0 such that f is a bijective map from U onto U' with C^1 inverse f^{-1}. A computation similar to the one in the proof of Proposition 1.4 shows that f^{-1} is also holomorphic. □

From the above, we have the following. Although the proofs are the same as the real case, we give them for the sake of completeness.

Theorem 1.5 (Implicit mapping theorem). *Let $f : (\mathbb{C}^n, 0) \to (\mathbb{C}^k, 0)$ be a holomorphic map. Suppose $n > k$ and 0 is a regular point of f. Renumbering the functions and the variables if necessary, we may assume*

$$\det \frac{\partial(f_1, \ldots, f_k)}{\partial(z_1, \ldots, z_k)}(0) \neq 0.$$

Then there exists a holomorphic map $\varphi : (\mathbb{C}^{n-k}, 0) \to (\mathbb{C}^k, 0)$ such that

$$f(\varphi_1(z_{k+1}, \ldots, z_n), \ldots, \varphi_k(z_{k+1}, \ldots, z_n), z_{k+1}, \ldots, z_n) = 0$$

for $(z_{k+1}, \ldots, z_n)$ in a neighborhood of 0.

Proof. Set $z' = (z_1, \ldots, z_k)$ and $z'' = (z_{k+1}, \ldots, z_n)$ and define a holomorphic map $g : (\mathbb{C}^n, 0) \to (\mathbb{C}^n, 0)$ by $g(z) = (f(z), z'')$. Then 0 is a regular point of g and thus g is biholomorphic by Theorem 1.4. We write $g^{-1} = (a, b)$ with $a : (\mathbb{C}^n, 0) \to (\mathbb{C}^k, 0)$ and $b : (\mathbb{C}^n, 0) \to (\mathbb{C}^{n-k}, 0)$. Then

$$f(a(z), b(z)) = z', \qquad b(z) = z''. \tag{1.4}$$

Thus if we set $\varphi(z'') = a(0, z'')$, φ is the desired holomorphic map. □

Remark 1.4. In the above situation, we may solve the equation

$$f(z_1, \ldots, z_k, \ldots, z_n) = 0$$

for $z_1, \ldots, z_k$ as functions $\varphi_1, \ldots, \varphi_k$ of $(z_{k+1}, \ldots, z_n)$ in a neighborhood of 0 and the set $f^{-1}(0)$ is the graph of the map $\varphi = (\varphi_1, \ldots, \varphi_k)$.

The following gives a local normal form of a map near a regular point:

Theorem 1.6. 1. *Let $f : (\mathbb{C}^d, 0) \to (\mathbb{C}^n, 0)$ be a holomorphic map. Suppose $d \leq n$ and 0 is a regular point of f. Setting $k = n - d$ and denoting by $w = (w_1 \ldots, w_d)$ coordinates on $\mathbb{C}^d$, we may assume that*

$$\det \frac{\partial(f_{k+1}, \ldots, f_n)}{\partial(w_1, \ldots, w_d)}(0) \neq 0.$$

Then there exists a biholomorphic map $h : (\mathbb{C}^n, 0) \to (\mathbb{C}^n, 0)$ such that

$$(h \circ f)(w_1, \ldots, w_d) = (0, \ldots, 0, w_1, \ldots, w_d).$$

2. *Let $f : (\mathbb{C}^n, 0) \to (\mathbb{C}^k, 0)$ be a holomorphic map. Suppose $n \geq k$ and 0 is a regular point of f. We may assume that*

$$\det \frac{\partial(f_1, \ldots, f_k)}{\partial(z_1, \ldots, z_k)}(0) \neq 0.$$

Then there exists a biholomorphic map $h : (\mathbb{C}^n, 0) \to (\mathbb{C}^n, 0)$ such that

$$(f \circ h)(z_1, \ldots, z_k, \ldots, z_n) = (z_1, \ldots, z_k).$$

Proof. **1.** We set $z' = (z_1, \ldots, z_k)$ and $z'' = (z_{k+1}, \ldots, z_n)$ and identify $\mathbb{C}^d$ with $\{0\} \times \mathbb{C}^d \subset \mathbb{C}^n$ by identifying w with $(0, z'')$, $z_{k+i} = w_i$. Define a map $g : (\mathbb{C}^n, 0) \to (\mathbb{C}^n, 0)$ by $g(z) = f(z'') + (z', 0)$. Then it is biholomorphic and $h = g^{-1}$ is the desired biholomorphic map.

2. If $n = k$, f is biholomorphic and thus set $h = f^{-1}$. If $n > k$, let $g : (\mathbb{C}^n, 0) \to (\mathbb{C}^n, 0)$ be as in the proof of Theorem 1.5 and set $h = g^{-1}$. Then from (1.4), we see that it is the desired map. $\square$

Remark 1.5. In the case 2 above, we may think of $(f_1, \ldots, f_k, z_{k+1}, \ldots, z_n)$ as a coordinate system on a neighborhood of 0 in $\mathbb{C}^n$.

The following theorems can be proved as in the one variable case or reducing to that case. We denote by D a connected open set in $\mathbb{C}^n$.

Theorem 1.7 (Uniqueness of analytic continuation). *Let f and g be holomorphic functions on D. If $f = g$ on a neighborhood of a point in D then $f = g$ on D.*

Theorem 1.8 (Maximum principle). *Let f be a holomorphic function on D. If there is a point z_0 in D such that $|f(z_0)| \geq |f(z)|$ for all z in a neighborhood of z_0, then f is constant on D.*

1.4 Germs of holomorphic functions

In this section we discuss basic local properties of holomorphic functions.

Let H denote the set of pairs (U, f) of a neighborhood U of 0 and a holomorphic function f on U. We define a relation $\sim$ in H as follows. For two elements (U, f) and (V, g) in H, we say $(U, f) \sim (V, g)$ if there is a neighborhood W of 0 such that $W \subset U \cap V$ and that the restrictions of f and g to W are the same. Then it is easily checked that $\sim$ is an equivalence relation in H. The equivalence class of (U, f) is called the *germ of f at* 0 and is denoted by $\boldsymbol{f}$, or simply by f if there is no fear of confusion.

We let $\mathscr{O}_n$ denote the quotient set $H/\sim$. The set $\mathscr{O}_n$ has the structure of a commutative ring with respect to the operations induced from the addition and multiplication of functions. It has the unity which is the equivalence class of the function constantly equal to 1. We call $\mathscr{O}_n$ the *ring of germs of holomorphic functions at* 0. In fact it contains the field $\mathbb{C}$ as the germs of constant functions and has a natural $\mathbb{C}$-algebra structure.

Remark 1.6. In algebraic terms the ring $\mathscr{O}_n$ is expressed as follows (cf. Section A.1). We may introduce an order relation in the set of neighborhoods of 0 by reverse inclusion, i.e., $U \leq V$ if $V \subset U$. Then it becomes a directed set. For a neighborhood U we denote by O_U the ring of holomorphic functions on U and for $V \subset U$ let $r_{VU} : O_U \to O_V$ be the restriction. Then (O_U, r_{VU}) is a direct system (cf. Definition A.8) and we may write

$$\mathscr{O}_n = \varinjlim_{U \ni 0} O_U,$$

the direct limit on the set of neighborhoods of 0.

If we denote by $\mathbb{C}\{z_1, \ldots, z_n\}$ the set of power series that converge absolutely in some neighborhood of 0, this set also has the structure of a ring. Since $(U, f) \sim (V, g)$ if and only if f and g have the same power series expansion, we may identify $\mathscr{O}_n$ with $\mathbb{C}\{z_1, \ldots, z_n\}$.

We start with some fundamental properties of the ring $\mathscr{O}_n$. For terminologies and basics of commutative algebra, we refer to Appendix A. In the following, we adopt the convention that $\mathbb{C}^0 = \{0\}$ and $\mathscr{O}_0 = \mathbb{C}$.

Proposition 1.5. *The ring $\mathscr{O}_n$ is an integral domain.*

Proof. Suppose $\boldsymbol{fg} = 0$ for germs $\boldsymbol{f}$ and $\boldsymbol{g}$ in $\mathscr{O}_n$. Then there is a connected neighborhood U of 0 such that $f(z)g(z) = 0$ for all z in U. If $\boldsymbol{f} \neq 0$, there is a point z such that $f(z) \neq 0$. By the continuity of f, we may find a neighborhood U_z of z such that $f(z') \neq 0$ for all z' in U_z. Since the restriction of g to U_z must be identically equal to 0, we get $\boldsymbol{g} = 0$ by Theorem 1.7. □

By the above proposition, we may form the fraction field of $\mathscr{O}_n$, which we denote by $\mathscr{M}_n$. Each element in $\mathscr{M}_n$ is expressed as $\boldsymbol{f}/\boldsymbol{g}$ and two expressions $\boldsymbol{f}/\boldsymbol{g}$ and $\boldsymbol{f}'/\boldsymbol{g}'$ stand for the same element if and only if $\boldsymbol{f}\boldsymbol{g}' = \boldsymbol{f}'\boldsymbol{g}$. We call an element of $\mathscr{M}_n$ a germ of a *meromorphic function* at 0 in $\mathbb{C}^n$.

Proposition 1.6. *A germ $\boldsymbol{u}$ in $\mathscr{O}_n$ is a unit if and only if it is the germ of a function u with $u(0) \neq 0$.*

Proof. If u is a holomorphic function in a neighborhood of 0 with $u(0) \neq 0$, then $1/u$ is a holomorphic function in a neighborhood of 0. Thus the germ of u is a unit in $\mathscr{O}_n$. The converse is obvious. □

If we denote by $\mathfrak{m}$ the set of non-units in $\mathscr{O}_n$, it is an ideal in $\mathscr{O}_n$.

Proposition 1.7. *The ideal $\mathfrak{m}$ is the unique maximal ideal in $\mathscr{O}_n$.*

Proof. Let I be an arbitrary ideal in $\mathscr{O}_n$. If I contains a unit, then $I = \mathscr{O}_n$. If I does not contain units, then $I \subset \mathfrak{m}$. □

Note that the quotient $\mathscr{O}_n/\mathfrak{m}$ is canonically isomorphic with $\mathbb{C}$. The isomorphism is given by assigning to the class of $\boldsymbol{f}$ the value $f(0)$. Recall that a ring with a unique maximal ideal is called a *local ring* (cf. Section A.2).

We analyze the structure of the ring $\mathscr{O}_n$ by induction on n. For this, we pay attention to a specific variable, which will be z_n. First, for a germ $\boldsymbol{f}$ in $\mathscr{O}_n$, we write $f = \sum_{|\nu|\geq 0} a_\nu z^\nu$.

Definition 1.10. If $\boldsymbol{f} \neq 0$, the *order* of $\boldsymbol{f}$ is the integer m such that $a_\nu = 0$ for all ν with $|\nu| < m$ and $a_{\nu_0} \neq 0$ for some ν_0 with $|\nu_0| = m$. The order of the germ 0 is defined to be $+\infty$.

Thus the order of $\boldsymbol{f}$ is a positive integer if and only if it is not 0 or a unit. The above notion of order does not depend on the coordinates, while the next one does.

Definition 1.11. The order of $\boldsymbol{f}$ in z_n is the order of $f(0, \ldots, 0, z_n)$ as a power series in z_n. We say that $\boldsymbol{f}$ is *regular in z_n of order m*, if the order m of $\boldsymbol{f}$ in z_n is a positive integer.

Thus $\boldsymbol{f}$ is regular in z_n of order m if and only if $f(0, \ldots, 0, z_n)$ has a zero of multiplicity m at $z_n = 0$.

Lemma 1.2. *If the order of $\boldsymbol{f}$ is m, then we may find a suitable coordinate system $(\zeta_1, \ldots, \zeta_n)$ of $\mathbb{C}^n$ such that the order of $\boldsymbol{f}$ in ζ_n is m.*

Proof. As the case $\boldsymbol{f} = 0$ is obvious, we assume that $\boldsymbol{f} \neq 0$. If we set $f_m = \sum_{|\nu|=m} a_\nu z^\nu$, then $f_m \neq 0$. Thus, there is a point $c = (c_1, \ldots, c_n) \neq 0$ such that $f_m(c) \neq 0$. There exists an $n \times n$ non-singular matrix C whose n-th row is c. If we set $\zeta = zC^{-1}$, $f(\zeta C)$ is of order m in ζ. Since $f_m((0, \ldots, 0, 1)C) = f_m(c) \neq 0$, we see that $f(\zeta C)$ is of order m in ζ_n. $\square$

Remark 1.7. The above arguments work for a finitely many germs in $\mathscr{O}_n$. Thus, a finite number of germs in $\mathscr{O}_n$ whose orders are positive integers can be simultaneously made to be regular in some variable, by a suitable change of coordinates.

We show that the ring $\mathscr{O}_n$ possesses certain properties similar to the ones polynomial rings do. Thus we set $z' = (z_1, \ldots, z_{n-1})$ and let $\mathscr{O}_{n-1}$ denote the ring of germs of holomorphic functions in z'. We consider the ring $\mathscr{O}_{n-1}[z_n]$ of polynomials in z_n with coefficients in $\mathscr{O}_{n-1}$:

$$\mathscr{O}_{n-1}[z_n] = \{\, \boldsymbol{a}_0 + \boldsymbol{a}_1 z_n + \cdots + \boldsymbol{a}_r z_n^r \;\mid r \in \mathbb{N}_0, \quad \boldsymbol{a}_i \in \mathscr{O}_{n-1} \,\},$$

which is considered as a subring of $\mathscr{O}_n$ and contains $\mathscr{O}_{n-1}$ as a subring.

Definition 1.12. Let m be a positive integer. A *Weierstrass polynomial* in z_n of degree m is a germ $\boldsymbol{h}$ in $\mathscr{O}_{n-1}[z_n]$ of the form

$$\boldsymbol{h} = \boldsymbol{a}_0 + \boldsymbol{a}_1 z_n + \cdots + \boldsymbol{a}_{m-1} z_n^{m-1} + z_n^m,$$

where $\boldsymbol{a}_0, \boldsymbol{a}_1, \ldots, \boldsymbol{a}_{m-1}$ are non-units in $\mathscr{O}_{n-1}$.

Thus a Weierstrass polynomial in z_n is a non-unit in both $\mathscr{O}_{n-1}[z_n]$ and $\mathscr{O}_n$. Note that in the above, $h(0, \ldots, 0, z_n) = z_n^m$. Hence $\boldsymbol{h}$ is regular in z_n of order m. In general, for every germ $\boldsymbol{f}$ in $\mathscr{O}_n$, we may write in a neighborhood of 0

$$f(z) = a_0 + a_1 z_n + \cdots + a_m z_n^m + \cdots$$

with a_i holomorphic functions in z'. The germ $\boldsymbol{f}$ is regular in z_n of order m if and only if $\boldsymbol{a}_0, \boldsymbol{a}_1, \ldots, \boldsymbol{a}_{m-1}$ are non-units in $\mathscr{O}_{n-1}$ and $\boldsymbol{a}_m$ is a unit in $\mathscr{O}_{n-1}$. In this case, $\boldsymbol{a}_m^{-1}(\boldsymbol{a}_0 + \boldsymbol{a}_1 z_n + \cdots + \boldsymbol{a}_m z_n^m)$ is a Weierstrass polynomial in z_n of degree m. The Weierstrass preparation theorem below shows that such a germ $\boldsymbol{f}$ is essentially equal to a Weierstrass polynomial of degree m.

We first prove the following:

Theorem 1.9 (Weierstrass division theorem). *Let* $\boldsymbol{h}$ *be a Weierstrass polynomial in* z_n *of degree* m. *Then, for every germ* $\boldsymbol{f}$ *in* $\mathscr{O}_n$, *there exist uniquely determined germs* $\boldsymbol{q}$ *in* $\mathscr{O}_n$ *and* $\boldsymbol{r}$ *in* $\mathscr{O}_{n-1}[z_n]$ *with* $\deg \boldsymbol{r} < m$ *such that*

$$\boldsymbol{f} = \boldsymbol{q}\boldsymbol{h} + \boldsymbol{r}.$$

Moreover, if $\boldsymbol{f}$ *is in* $\mathscr{O}_{n-1}[z_n]$, *so is* $\boldsymbol{q}$.

Proof. We choose $\rho = (\rho_1, \dots, \rho_n)$, $\rho_i > 0$, small enough so that the germs $\boldsymbol{f}$ and $\boldsymbol{h}$ have representatives on an open set containing $\overline{\Delta(0,\rho)}$. As $\boldsymbol{h}$ is regular in z_n, taking a smaller ρ if necessary, we may assume that $h(z) \neq 0$ for $z = (z_1, \dots, z_n)$ with $|z_1| < \rho_1, \dots, |z_{n-1}| < \rho_{n-1}, |z_n| = \rho_n$. We denote by γ the path in the z_n- plane that turns counterclockwise along the circle $|z_n| = \rho_n$. If we set

$$q(z) = \frac{1}{2\pi\sqrt{-1}} \int_\gamma \frac{f(z',\zeta)}{h(z',\zeta)} \frac{d\zeta}{\zeta - z_n},$$

then $q(z)$ is holomorphic in $\Delta(0,\rho)$. If we further set $r(z) = f(z) - q(z)h(z)$, using the Cauchy integral formula

$$f(z) = \frac{1}{2\pi\sqrt{-1}} \int_\gamma \frac{f(z',\zeta)}{\zeta - z_n} d\zeta,$$

we compute

$$r(z) = \frac{1}{2\pi\sqrt{-1}} \int_\gamma \frac{f(z',\zeta)}{h(z',\zeta)} \frac{h(z',\zeta) - h(z',z_n)}{\zeta - z_n} d\zeta.$$

The first function in the integrant does not depend on z_n and the second is a polynomial in z_n of degree less than m, since $\boldsymbol{h}$ is a Weierstrass polynomial in z_n of degree m. Thus we see that $\boldsymbol{r}$ is in $\mathscr{O}_{n-1}[z_n]$ and $\deg \boldsymbol{r} < m$.

To show the uniqueness of $\boldsymbol{q}$ and $\boldsymbol{r}$, suppose $\boldsymbol{f} = \boldsymbol{q}\boldsymbol{h} + \boldsymbol{r} = \boldsymbol{q}'\boldsymbol{h} + \boldsymbol{r}'$. Choose ρ as above. Then we have

$$r'(z',z_n) - r(z',z_n) = (q(z',z_n) - q'(z',z_n))h(z',z_n)$$

in $\Delta(\rho, 0)$. For each fixed a' in $\Delta(\rho', 0)$, $\rho' = (\rho_1, \dots, \rho_{n-1})$, $h(a', z_n)$ has m zeros in $\Delta(\rho_n, 0)$, counting multiplicities. On the other hand, the left-hand side above is a polynomial in z_n of degree at most $m-1$ and thus $r'(a', z_n) - r(a', z_n) = 0$ in $\Delta(\rho_n, 0)$. Therefore $\boldsymbol{r}' = \boldsymbol{r}$ and also $\boldsymbol{q}' = \boldsymbol{q}$, as $\mathscr{O}_n$ is an integral domain.

The last statement is the division algorithm for polynomials. $\square$

Theorem 1.10 (Weierstrass preparation theorem). *Let $\boldsymbol{f}$ be a germ in $\mathscr{O}_n$ which is regular in z_n of order m. Then there is a unique Weierstrass polynomial $\boldsymbol{h}$ in z_n of degree m such that $\boldsymbol{f} = \boldsymbol{uh}$ with $\boldsymbol{u}$ a unit in $\mathscr{O}_n$.*

Proof. We introduce new variables $w = (w_1, \dots, w_m)$ and we set $\mathscr{O}_{m+n} = \mathbb{C}\{w, z\}$ and $\mathscr{O}_{m+n-1} = \mathbb{C}\{w, z'\}$. We set

$$p(w, z_n) = w_m + w_{m-1} z_n + \cdots + w_1 z_n^{m-1} + z_n^m$$

and think of it as a Weierstrass polynomial in $\mathscr{O}_{m+n-1}[z_n]$ of degree m. Since $\mathscr{O}_n$ is naturally a subring of $\mathscr{O}_{m+n}$, by the Weierstrass division theorem, there exist unique germs $\boldsymbol{q}$ in $\mathscr{O}_{m+n}$ and $\boldsymbol{r}$ in $\mathscr{O}_{m+n-1}[z_n]$ of degree less than m such that

$$\boldsymbol{f} = \boldsymbol{qp} + \boldsymbol{r}. \tag{1.5}$$

We write

$$r(w, z', z_n) = b_m(w, z') + b_{m-1}(w, z') z_n + \cdots + b_1(w, z') z_n^{m-1}$$

with $\boldsymbol{b}_1, \dots, \boldsymbol{b}_m$ in $\mathscr{O}_{m+n-1}$. If we set $w = 0$ and $z' = 0$ in (1.5), we get

$$f(0, z_n) = q(0, 0, z_n) z_n^m + b_m(0,0) + b_{m-1}(0,0) z_n + \cdots + b_1(0,0) z_n^{m-1}.$$

Since f is of order m in z_n, $q(0,0,0) = c \neq 0$ and

$$b_1(0,0) = \cdots = b_{m-1}(0,0) = b_m(0,0) = 0.$$

If we differentiate the both sides of (1.5) with respect to w_j, noting that $\partial p / \partial w_j = z_n^{m-j}$, we get

$$0 = \frac{\partial q}{\partial w_j} p + q z_n^{m-j} + \frac{\partial b_m}{\partial w_j} + \frac{\partial b_{m-1}}{\partial w_j} z_n + \cdots + \frac{\partial b_1}{\partial w_j} z_n^{m-1}.$$

Setting $w = 0$ and $z' = 0$ again, we see that

$$\frac{\partial b_i}{\partial w_j}(0,0) = \begin{cases} -c & \text{for } \ i = j, \\ 0 & \text{for } \ i > j. \end{cases}$$

Thus

$$\det \frac{\partial(b_1, \dots, b_m)}{\partial(w_1, \dots, w_m)}(0,0) = (-c)^m \neq 0$$

and by Theorem 1.5, there is a holomorphic map $\varphi : (\mathbb{C}^{n-1}, 0) \to (\mathbb{C}^m, 0)$ such that

$$b_1(\varphi(z'), z') = \cdots = b_{m-1}(\varphi(z'), z') = b_m(\varphi(z'), z') = 0.$$

If we substitute $w = \varphi(z')$ in (1.5), we get

$$f(z', z_n) = q(\varphi(z'), z', z_n) p(\varphi(z'), z_n).$$

If we set $u(z', z_n) = q(\varphi(z'), z', z_n)$ and $h(z', z_n) = p(\varphi(z'), z_n)$, then $\boldsymbol{f} = \boldsymbol{uh}$ with $\boldsymbol{u}$ a unit in $\mathscr{O}_n$ and $\boldsymbol{h}$ a Weierstrass polynomial in z_n of degree m. The uniqueness of such $\boldsymbol{u}$ and $\boldsymbol{h}$ is not difficult to see. $\square$

Remark 1.8. By the above theorem and Remark 1.7, for a finite number of germs, each of which is not 0 or a unit in $\mathscr{O}_n$, we may find a coordinate system $(z_1, \dots, z_n)$ such that each germ is expressed as the product of a unit and a Weierstrass polynomial in z_n.

Next we discuss some important properties of the ring $\mathscr{O}_n$ which follow from the above theorems. First we observe that:

Lemma 1.3. *Let* $\boldsymbol{h}$ *be a Weierstrass polynomial in* $\mathscr{O}_{n-1}[z_n]$. *Then* $\boldsymbol{h}$ *is irreducible in* $\mathscr{O}_{n-1}[z_n]$ *if and only if it is irreducible in* $\mathscr{O}_n$. *If it is not irreducible, all of its factors are Weierstrass polynomials in* $\mathscr{O}_{n-1}[z_n]$.

Proof. Suppose $\boldsymbol{h}$ is not irreducible in $\mathscr{O}_n$. Then we may write $\boldsymbol{h} = \boldsymbol{f}_1\boldsymbol{f}_2$ with $\boldsymbol{f}_i$, $i = 1, 2$, non-units in $\mathscr{O}_n$. Since $\boldsymbol{h}$ is regular in z_n, both $\boldsymbol{f}_1$ and $\boldsymbol{f}_2$ are regular in z_n and by the preparation theorem, we may write $\boldsymbol{f}_i = \boldsymbol{u}_i\boldsymbol{h}_i$, $i = 1, 2$, with $\boldsymbol{u}_i$ units in $\mathscr{O}_n$ and $\boldsymbol{h}_i$ Weierstrass polynomials in $\mathscr{O}_{n-1}[z_n]$. By the uniqueness, we have $\boldsymbol{h} = \boldsymbol{h}_1\boldsymbol{h}_2$, and thus $\boldsymbol{h}$ is not irreducible in $\mathscr{O}_{n-1}[z_n]$.

Conversely suppose $\boldsymbol{h}$ is not irreducible in $\mathscr{O}_{n-1}[z_n]$. Then $\boldsymbol{h} = \boldsymbol{g}_1\boldsymbol{g}_2$ with $\boldsymbol{g}_i$, $i = 1, 2$, non-units in $\mathscr{O}_{n-1}[z_n]$. If one of $\boldsymbol{g}_1$ or $\boldsymbol{g}_2$, say $\boldsymbol{g}_1$, were a unit in $\mathscr{O}_n$, we would have $\boldsymbol{u}\boldsymbol{g}_1 = 1$ for some $\boldsymbol{u}$ in $\mathscr{O}_n$ and $\boldsymbol{g}_2 = \boldsymbol{u}\boldsymbol{h}$. From the last part of the division theorem, we would then see that $\boldsymbol{u} \in \mathscr{O}_{n-1}[z_n]$, which contradicts the fact that $\boldsymbol{g}_1$ is a non-unit in $\mathscr{O}_{n-1}[z_n]$. Thus $\boldsymbol{g}_1$ and $\boldsymbol{g}_2$ are non-units in $\mathscr{O}_n$ and $\boldsymbol{h}$ is not irreducible in $\mathscr{O}_n$. $\square$

Theorem 1.11. *The ring* $\mathscr{O}_n$ *is a unique factorization domain.*

Proof. We proceed by induction on n. First, we have $\mathscr{O}_0 = \mathbb{C}$, which is a unique factorization domain, a UFD for short. We assume that $\mathscr{O}_{n-1}$ is a UFD. Take a germ $\boldsymbol{f}$ in $\mathscr{O}_n$ which is not 0 or a unit. By changing the coordinate system on $\mathbb{C}^n$ if necessary, we may assume that $\boldsymbol{f}$ is regular in z_n. By the Weierstrass preparation theorem, there is a unit $\boldsymbol{u}$ in $\mathscr{O}_n$ such that $\boldsymbol{u}\boldsymbol{f}$ is a Weierstrass polynomial in $\mathscr{O}_{n-1}[z_n]$, which is a UFD by the induction hypothesis and Gauss' lemma (Theorem A.15). Since $\boldsymbol{u}\boldsymbol{f}$ is not 0 or a unit, it can be expressed as a product of irreducible elements in $\mathscr{O}_{n-1}[z_n]$. Then the theorem follows from Lemma 1.3. $\square$

Since a unit in $\mathscr{O}_n$ has m-th roots for every positive integer m, every germ $\boldsymbol{f}$, which is not 0 or a unit, has the irreducible decomposition (cf. (A.11)) of the form

$$\boldsymbol{f} = \boldsymbol{p}_1^{m_1} \cdots \boldsymbol{p}_r^{m_r}.$$

Recall that, in a UFD, an element is irreducible if and only if it is prime.

Theorem 1.12. *The ring $\mathscr{O}_n$ is Noetherian.*

Proof. We proceed by induction on n. We have $\mathscr{O}_0 = \mathbb{C}$, which is Noetherian. We assume that $\mathscr{O}_{n-1}$ is Noetherian.

Take an arbitrary ideal I in $\mathscr{O}_n$. If $I = 0$, we have nothing to prove. If $I \neq 0$, we choose a non-zero element $\boldsymbol{h}$ in I. By the Weierstrass preparation theorem, we may assume that $\boldsymbol{h}$ is a Weierstrass polynomial. Thus $\boldsymbol{h}$ is in $I \cap \mathscr{O}_{n-1}[z_n]$. Since $I \cap \mathscr{O}_{n-1}[z_n]$ is an ideal in $\mathscr{O}_{n-1}[z_n]$, which is Noetherian by the induction hypothesis and the Hilbert basis theorem (Theorem A.17), it has a finite number of generators $\boldsymbol{g}_1, \ldots, \boldsymbol{g}_r$. We show that they also generate the ideal I over $\mathscr{O}_n$. Take an element $\boldsymbol{f}$ in I. Changing the coordinate system of $\mathbb{C}^n$ if necessary, we may assume that $\boldsymbol{f}$ is regular in z_n. Then by the Weierstrass division theorem, we may write $\boldsymbol{f} = \boldsymbol{q}\boldsymbol{h} + \boldsymbol{r}$ for some $\boldsymbol{q}$ in $\mathscr{O}_n$ and $\boldsymbol{r}$ in $\mathscr{O}_{n-1}[z_n]$. Since the germs $\boldsymbol{f}$ and $\boldsymbol{h}$ are in I, so is $\boldsymbol{r}$. Thus $\boldsymbol{r}$ is in $I \cap \mathscr{O}_{n-1}[z_n]$. Since $\boldsymbol{h}$ and $\boldsymbol{r}$ can be written as linear combinations of $\boldsymbol{g}_1, \ldots, \boldsymbol{g}_r$ over $\mathscr{O}_{n-1}[z_n]$, the germ $\boldsymbol{f}$ is a linear combination of $\boldsymbol{g}_1, \ldots, \boldsymbol{g}_r$ over $\mathscr{O}_n$. $\square$

We give another application of Theorem 1.10. For a point z in $\mathbb{C}^n$, let $\mathscr{O}_{n,z}$ be the ring of germs of holomorphic functions at z, which is naturally isomorphic with $\mathscr{O}_n$. For a germ $\boldsymbol{f}$ in $\mathscr{O}_n$ we denote by f_z the germ of f at z near 0.

Proposition 1.8. *If $\boldsymbol{f}$ and $\boldsymbol{g}$ are relatively prime in $\mathscr{O}_n$, then f_z and g_z are relatively prime in $\mathscr{O}_{n,z}$ for all z sufficiently close to* 0.

Proof. If $\boldsymbol{f}$ or $\boldsymbol{g}$ is a unit, the proposition obviously holds. Thus we assume that $\boldsymbol{f}$ and $\boldsymbol{g}$ are non-units. By the preparation theorem, we may assume that they are Weierstrass polynomials in z_n (cf. Remark 1.8) and are relatively prime in $\mathscr{O}_{n-1}[z_n]$ (cf. Lemma 1.3). Thus by Theorem A.16, there exist elements $\boldsymbol{a}$ and $\boldsymbol{b}$ in $\mathscr{O}_{n-1}[z_n]$ and $\boldsymbol{c} \neq 0$ in $\mathscr{O}_{n-1}$ such that

$$\boldsymbol{a}\boldsymbol{f} + \boldsymbol{b}\boldsymbol{g} = \boldsymbol{c}. \tag{1.6}$$

Note that the above equality holds in a neighborhood U of 0 in $\mathbb{C}^n$ where all the germs involved have representatives.

We may assume that U is of the form $U = U' \times U''$ with U' a neighborhood of 0 in $\mathbb{C}^{n-1}$ and U'' a neighborhood of 0 in $\mathbb{C} = \{z_n\}$. As $\boldsymbol{f}$ is regular in z_n, taking a smaller U' if necessary, we may assume that for each z' in U', $f(z', z_n)$ is not identically zero as a function of z_n in U''.

Let a be a point in U. If $f(a) \neq 0$ or $g(a) \neq 0$, f_a and g_a are relatively prime in $\mathscr{O}_{n,a}$. Thus assume that $f(a) = g(a) = 0$. If f_a and g_a have a common factor h_a in $\mathscr{O}_{n,a}$ with $h(a) = 0$, h_a also divides c_a by (1.6) so that the function $h(z', z_n)$ defined near $a = (a', a_n)$ depends only on z'. Thus $h(a', z_n)$ vanishes identically in z_n. This contradicts the fact that $f(a', z_n)$ is not identically zero as a function of z_n. $\square$

We finish this section by discussing the *Riemann extension theorem* in the several variable case.

Theorem 1.13. *Let D be a connected open set in $\mathbb{C}^n$ and g a function holomorphic and not identically 0 in D. We set $V = \{ z \in D \mid g(z) = 0 \}$. If f is a function holomorphic in $D \setminus V$ and is locally bounded, i.e., each point z in D has a neighborhood U such that f is bounded in $(D \setminus V) \cap U$, there is a unique function $\tilde{f}$ holomorphic in D and equal to f on $D \setminus V$.*

Proof. The case $n = 1$ is the classical Riemann extension theorem. Thus we assume that $n \geq 2$ and reduce the problem to this case.

First we note that, by the assumption and Theorem 1.7, V has no interior points. In particular, for every point z in V, the germ g_z is not 0 or a unit. For the existence, it suffices to show that each point z in V has a neighborhood on which there is an extension $\tilde{f}$ of f. Without loss of generality, we may assume $z = 0$. After a change of coordinates if necessary, we may assume that $\boldsymbol{g}$ is regular in z_n, in fact we may assume that $\boldsymbol{g}$ is a Weierstrass polynomial in z_n. We choose $\rho = (\rho_1, \ldots, \rho_n)$, $\rho_i > 0$, small enough so that $\boldsymbol{g}$ has a representative on an open set containing $\overline{\Delta(0, \rho)}$ and that $g(z) \neq 0$ for $z = (z', z_n)$ with $z' \in \Delta(0, \rho')$ and $|z_n| = \rho_n$, where we set $z' = (z_1, \ldots, z_{n-1})$ and $\rho' = (\rho_1, \ldots, \rho_{n-1})$. Denoting by γ the path in the z_n-plane turning counterclockwise along the circle $|z_n| = \rho_n$, we set

$$\tilde{f}(z) = \frac{1}{2\pi\sqrt{-1}} \int_\gamma \frac{f(z', \zeta)}{\zeta - z_n} d\zeta.$$

Then $\tilde{f}$ is holomorphic in $\Delta(0, \rho)$. For each fixed z' in $\Delta(0, \rho')$, $f(z', z_n)$ is holomorphic in z_n in an open set U containing $\overline{\Delta(0, \rho_n)}$ except for a finite number of points (the zeros of $g(z', z_n)$) and is locally bounded. Thus by the Riemann extension theorem in one variable case, $f(z', z_n)$ has a holomorphic extension in U. Therefore by the Cauchy integral formula, $\tilde{f}$ is an extension of f in $\Delta(0, \rho)$. The uniqueness follows from the uniqueness of analytic continuation. $\square$

Corollary 1.1. *If D and V are as in Theorem 1.13, $D \setminus V$ is connected.*

Proof. If $D \setminus V$ is not connected, there are non-empty open sets U_1 and U_2 such that $D \setminus V = U_1 \cup U_2$ and $U_1 \cap U_2 = \emptyset$. The function f, defined by $f(z) = 1$ for $z \in U_1$ and $f(z) = 0$ for $z \in U_2$, has no extensions. $\square$

Notes

For Section 1.2, there are a number of textbooks. Here we list [Cartan (1961)]. See 15.12 Theorem in [Rudin (1987)] for Remark 1.3.1.

We list [Fritzsche and Grauert (2002); Gunning and Rossi (1965)] as general references for the theory of analytic functions of several complex variables. The proof of the Weierstrass preparation theorem (Theorem 1.10) given here is based on the one in [Noguchi and Fukuda (1976)].